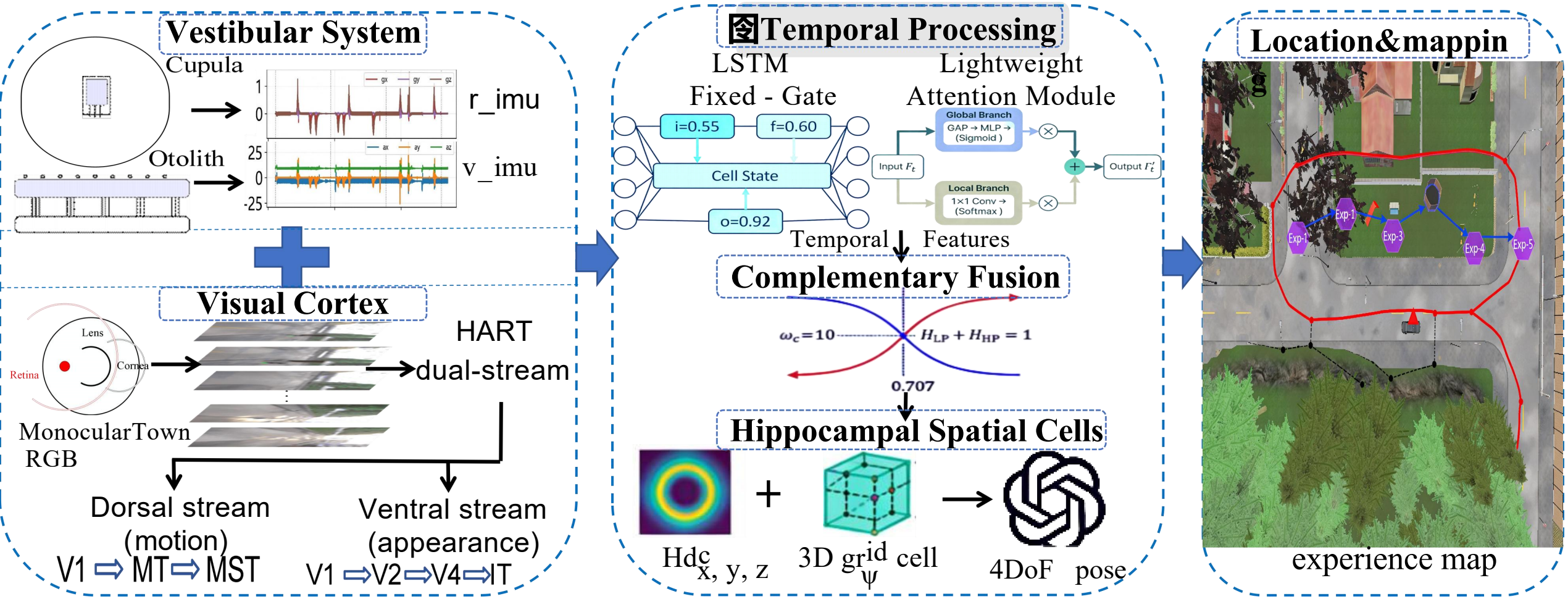
Vestibular System
Cupula
Otolith
r_imu
v_imu
Visual Cortex
Lens
Cornea
Retina
HART
dual-stream
MonocularTown
RGB
Dorsal stream
(motion)
V1 ⇨ MT ⇨ MST
Ventral stream
(appearance)
V1 ⇨ V2 ⇨ V4 ⇨ IT
Temporal Processing
LSTM
Fixed - Gate
i=0.55
f=0.60
Cell State
o=0.92
Lightweight
Attention Module
Global Branch
GAP → MLP →
(Sigmoid)
Input F_t
Output F'_t
Local Branch
1×1 Conv →
(Softmax)
Temporal Features
Complementary Fusion
$\omega_c = 10$
$H_{LP} + H_{HP} = 1$
0.707
Hippocampal Spatial Cells
Hdc
x, y, z
3D grid cell
ψ
4DoF pose
Location&mappin
Exp-1
Exp-3
Exp-4
Exp-5
experience map